

llmXive Automated Review of EnterpriseClawBench: Benchmarking Agents from Real Workplace Sessions

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (8 lenses): Claim Accuracy, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper makes several strong causal and methodological claims that are not fully supported by the cited sources or the presented data.

First, in Section 3.1, the authors attribute their specific five-dimension semantic rubric (grounded accuracy, task relevance, substantive depth, practical utility, communication quality) to “multi-metric LLM-evaluation practice” citing Helm, G-Eval, and RAGAS. While these works are seminal in the field, they do not explicitly propose or validate this specific 5-point taxonomy for *enterprise artifact* evaluation. The claim implies a direct derivation that is not present in the cited literature; the rubric appears to be an ad-hoc construction by the authors rather than a standard practice established by these references.

Second, in Section 3.2, the explanation for the “Claude-family drop under Hermes” relies on the claim that “Trace inspection suggests a runtime-level mismatch.” The paper asserts that Hermes blocks active probing or truncates traces, causing failures. However, the text provides no quantitative evidence from the traces (e.g., counts of blocked tool calls, truncation events, or specific error logs) to substantiate this mechanism. The claim is presented as a fact derived from inspection, but without the supporting data or a reference to a specific trace analysis table/figure, it remains an unverified hypothesis.

Third, regarding the “Role-class effects” in Section 3.2, the authors claim that manual inspection suggests marketing and finance tasks are harder due to “heavy document comprehension” and “company-specific scenarios.” While plausible, this is presented as an explanatory finding for the observed score gaps without statistical backing (e.g., a correlation between document length/complexity and score, or a controlled ablation). The claim overstates the evidence, which is currently limited to a qualitative observation.

These issues do not invalidate the benchmark’s utility but require the authors to either provide the missing quantitative evidence for their causal claims or soften the language to reflect that these are hypotheses requiring further validation.

Action items.

- **[writing]** Section 3.1 cites Helm/G-Eval/RAGAS to support a specific 5-dimension rubric for enterprise artifacts. These sources do not define this specific taxonomy; the claim overstates the direct lineage of the rubric design from these general frameworks.
- **[writing]** Section 3.2 claims ‘Trace inspection suggests a runtime-level mismatch’ explains Claude/Hermes failures. No quantitative evidence (e.g., block rates, truncation counts) is provided to support this causal mechanism, leaving it as an unverified hypothesis.
- **[writing]** Section 3.2 attributes lower scores in marketing/finance to ‘heavy document comprehension’ based on ‘manual inspection.’ This causal link lacks statistical correlation or controlled analysis, presenting a hypothesis as a supported finding.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on specialized terminology and internal project shorthand that excludes non-specialist readers. While the authors define some terms (like ‘session’), several key concepts are introduced without sufficient plain-language explanation.

First, the term ‘**claws**’ is introduced in the Introduction as a synonym for ‘harnesses’ following a ‘recent line of Claw-style benchmarks.’ However, the text does not explicitly state that ‘claw’ is a shorthand for ‘agent harness’ before using it as a standalone noun. This creates a barrier for readers unfamiliar with the specific ‘Claw’ ecosystem. The phrase ‘we call such harnesses claws’ is a definition, but it is buried in a complex sentence. A clearer, standalone definition is needed.

Second, the term ‘**fixtures**’ is used repeatedly (Abstract, Section 2) to refer to input files or data dependencies required for a task. In software testing, ‘fixtures’ is standard, but in the context of general AI agent evaluation, ‘input files’ or ‘required data’ is more accessible. The text assumes the reader understands that ‘recovered fixtures’ means ‘recovered input files.’

Third, ‘**TaskInstances**’ (Section 2, ‘Benchmark statistics’) is capitalized and used as a proper noun without definition. It appears to refer to raw, unprocessed task candidates from the session archive. Defining this as ‘raw task candidates’ or ‘initial task records’ would improve clarity.

Fourth, the abbreviation ‘**hm**’ (for ‘harness–model’) is defined via a LaTeX macro but the concept of the ‘harness–model combination’ is central to the paper’s argument. The text often refers to ‘hm combinations’ or ‘hm performance’ without ensuring the reader fully grasps that this refers to the specific pairing of an agent framework (harness) and a language model.

Finally, terms like ‘**rubrics**’ (Abstract, Section 2) and ‘**skill subclasses**’ (Section 2) are used without plain-English equivalents. ‘Rubrics’ should be clarified as ‘scoring criteria’ or ‘evaluation guidelines,’ and ‘skill subclasses’ should be explained as ‘specific types of tasks within a broader role category.’

These issues, while fixable with minor edits, currently create a jargon-heavy barrier that limits the paper’s accessibility to a broader audience beyond the immediate ‘Claw’ research community.

Action items.

- **[writing]** The manuscript relies heavily on specialized terminology and internal project shorthand that excludes non-specialist readers. While the authors define some terms (like ‘session’), several key concepts are introduced without sufficient plain-language explanation. First, the term ‘claws’ is introduced in the Introduction as a synonym for ‘harnesses’ following a ‘recent line of Claw-style benchmarks.’ However, the text does not explicitly state that ‘claw’ is a shorthand for ‘agent harness’ before

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper generally maintains a strong logical flow from the problem statement (enterprise agent evaluation gaps) to the proposed solution (EnterpriseClawBench) and the resulting evidence. The construction pipeline logic is sound: filtering noisy sessions for reproducibility (fixture recovery, network checks) logically precedes the creation of a benchmark, ensuring that failures are due to agent capability rather than data issues. The argument that “harness-model” combinations are a critical evaluation dimension is well-supported by the specific example of the Claude/Hermes mismatch, where the same model performs differently under different harnesses.

However, there are minor gaps in the logical support for specific causal claims and experimental design details. First, in the “Skill Evaluation” section, the text asserts that skills are distilled from 10 in-domain tasks and tested on 5 held-out tasks. While the term “held-out” implies a disjoint set, the text does not explicitly define the total size of the “frontend page generation” subclass. Without this context, the logical validity of the “held-out” claim (i.e., that the 5 tasks are truly independent of the 10 used for distillation) is not fully established. A brief mention of the subclass size or the sampling method would close this gap.

Second, the causal explanation for the “Harness-model interaction” results relies heavily on qualitative “trace inspection.” The authors claim that Hermes blocks actions or truncates traces, leading to lower scores. While this is a plausible mechanism, the logical link would be strengthened by a specific quantitative observation (e.g., “Hermes blocked X% of script execution attempts compared to Y% for Claude Code”) rather than a general description. The current phrasing suggests the conclusion is derived from the traces, but the evidence provided in the text is descriptive rather than evidentiary.

Finally, the “Judge Reliability” section notes a negative rank correlation for visual judges against humans. The logical conclusion that “evaluation on multimodal artifacts is not yet mature” is sound, but the paper could more explicitly connect the specific nature of the visual artifacts (e.g., spreadsheets with masked data) to the failure mode to strengthen the causal argument for *why* the visual judge fails, rather than just stating *that* it fails.

Action items.

- **[writing]** In the ‘Skill Evaluation’ section, the text claims the experiment uses ‘10 in-domain tasks’ for skill creation and ‘5 held-out tasks’ for evaluation. However, the text does not explicitly state the total number of tasks in the ‘frontend page generation’ subclass from which these were sampled. Without knowing the total pool size, the claim that the 5 tasks are ‘held-out’ (implying no overlap with the 10) lacks sufficient logical grounding regarding the sampling strategy.
- **[writing]** The ‘Harness-model interaction’ section attributes the performance drop of Claude models under Hermes to ‘runtime-level mismatch’ and ‘approval checks’. While the text states ‘Trace inspection suggests...’, it does not provide a specific quantitative metric (e.g., percentage of blocked calls, average trace truncation length) to logically support the causal link between the specific Hermes behavior and the observed score drop, relying instead on qualitative description.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper makes several claims that extend beyond the immediate evidence provided, particularly regarding the scope of the skill evaluation and the degree of automation in the construction pipeline.

First, the Abstract and Introduction claim that the benchmark offers “native support for evaluating skill generalization across held-out tasks from the same enterprise task class.” However, Section 5.3 (“Skill Evaluation”) reveals that this evaluation is conducted on a single task subclass (frontend page generation) involving only 15 tasks (10 for training, 5 for testing). While the results are interesting, extrapolating this to a general feature of the benchmark for “enterprise skills” is an overreach. The sample size is too small to support a general claim about the benchmark’s ability to evaluate skill transfer across the diverse 45 role-specific subclasses mentioned in Section 3. The authors should temper this claim to reflect that this is a pilot study or proof-of-concept rather than a fully realized evaluation framework for all skill types.

Second, the paper repeatedly describes the construction process as an “automated construction protocol” (Abstract, Introduction, Conclusion). While the pipeline includes automated gates, the Appendix and Section 3 reveal significant human involvement. The “Lite” set is explicitly “manually audited,” and the text notes “manual inspection” of traces to explain role-class performance differences. Furthermore, the redaction recovery and self-containment checks described in the Appendix involve heuristic rules that likely require human tuning or verification to ensure quality. Describing the entire process as “automated” without qualifying the extent of human curation and manual auditing overstates the system’s autonomy and reproducibility.

Finally, the claim that the benchmark represents “real workplace sessions” (Abstract) may be compromised by the aggressive filtering process. The funnel in Figure 2 shows a reduction from 5,291 raw instances to 852 tasks. The rejection criteria (e.g., network reachability, fixture recovery, self-containment) inherently filter out tasks that are ambiguous, broken, or dependent on external, unstable resources. While this ensures reproducibility, it risks creating a benchmark that is “cleaner” and more structured than the actual “real workplace” environment, potentially missing the very edge cases and ambiguities that characterize real enterprise agent failures. The paper should more honestly discuss how this filtering might bias the benchmark away from the full complexity of real-world sessions.

Action items.

- **[science]** The claim that the benchmark ‘naturally’ supports skill generalization overstates the evidence. The evaluation is restricted to a single subclass (frontend page generation) with only 15 total tasks. The paper should explicitly qualify this as a ‘proof-of-concept’ rather than a general capability of the benchmark, as the sample size is insufficient to support broad claims about skill transfer across the full 852-task set.

- **[writing]** The assertion that the pipeline ‘automatically converts’ sessions overclaims the reality. The Appendix details significant manual intervention, including ‘manual auditing’ of the 120-task Lite set and human-in-the-loop decisions for redaction recovery. The text should clarify the extent of human curation versus full automation to avoid misleading readers about the pipeline’s autonomy.
- **[science]** The claim that the benchmark represents ‘real workplace sessions’ may be compromised by aggressive filtering. Rejection criteria (e.g., network reachability, self-containment) may systematically exclude complex, ambiguous real-world tasks, creating a ‘cleaned’ benchmark that does not fully represent the ‘real workplace’ chaos it claims to model. This limitation needs stronger discussion.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The paper addresses safety and ethics primarily through the decision not to release the dataset due to proprietary and sensitive content (Abstract, Section 3, Limitations). The authors acknowledge the need for “redaction and recovery audit records” in the Ethical Considerations section. However, the current treatment of these issues is insufficient for a paper claiming to use real-world employee data.

First, the “Ethical Considerations” section (Section 7) is overly generic. It advises future users on what *should* be done but fails to document what *was* done. The manuscript mentions “privacy checks” and “redaction recovery” in the pipeline description (Section 3) but provides no technical or procedural details on how PII (Personally Identifiable Information), trade secrets, or sensitive business logic were removed from the 5,291 raw TaskInstances. The Appendix shows masked examples (e.g., ****), but the methodology for achieving this masking—whether automated, manual, or hybrid—and the error rates of this process are not described. To ensure the benchmark does not inadvertently leak sensitive information, the authors must detail the specific redaction mechanisms and the audit trail for the 852 final tasks.

Second, and more critically, the paper lacks any mention of Institutional Review Board (IRB) approval or informed consent regarding the source data. The data consists of “continuous internal use” by employees at an AI startup (Section 3). Using employee work interactions, even if anonymized, for public research and benchmarking typically requires ethical clearance. The authors must clarify whether this research was reviewed by an IRB or an equivalent ethics committee, or provide a strong justification for why such review was not required (e.g., if the data was fully de-identified and the research posed minimal risk). The absence of this information is a significant gap in the ethical reporting of the study.

Finally, the human evaluation component (Section 5.4, Table 4) involved 48 packets audited by humans. The paper does not state whether these human raters were informed of the sensitive

nature of the data, if they signed non-disclosure agreements (NDAs), or if they were compensated appropriately. Given the potential for exposure to sensitive business data during the audit, the safety protocols for the human raters should be explicitly described.

Action items.

- **[writing]** The ‘Ethical Considerations’ section (Section 7) is currently a single paragraph of general advice. It must be expanded to explicitly detail the specific redaction protocols used (e.g., regex patterns, manual audit steps), the criteria for ‘risk-marking’ instances, and the governance process for the 48 human-audited packets. Without these specifics, the claim of ‘privacy checks’ is unverifiable.
- **[writing]** The paper states the benchmark is built from ‘internal enterprise sessions’ of 100+ employees (Section 3) but does not mention IRB approval, informed consent, or employee notification. Even if data is anonymized, the use of employee work artifacts for public benchmarking requires explicit ethical clearance or a clear justification of why such review was waived. This must be addressed.

Scientific Evidence

Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The paper presents a compelling benchmark derived from real enterprise sessions, but the scientific evidence supporting the robustness of the evaluation protocol and the generalizability of the skill-transfer claims requires strengthening.

First, the **skill evaluation** (Section 5.3) is the most significant statistical weakness. The authors claim that “skill injection is therefore high variance” and that outcomes depend on creator-consumer fit. However, this conclusion is drawn from a single task subclass (frontend page generation) with a total of only 15 tasks (10 in-domain, 5 held-out). With such a small sample size, the observed variance could easily be driven by idiosyncrasies of the specific HTML generation task rather than a general property of skill transfer. The current evidence does not support a general claim about skill evaluation; it only supports a case study on frontend generation. The authors should either expand this experiment to include 2-3 additional task classes (e.g., spreadsheet analysis, document drafting) to demonstrate consistency or significantly temper their conclusions to reflect the limited scope.

Second, the **human judge calibration** (Section 5.4, Table 4) relies on a sample of 48 packets. While the text route correlation (Spearman $\rho = 0.790$) is encouraging, the visual route shows a negative rank correlation ($\rho = -0.259$). Given that the benchmark explicitly targets multimodal artifacts (spreadsheets, slides, HTML), the visual judge is a critical component of the evaluation. A sample of 24 visual artifacts is statistically underpowered to definitively characterize the failure mode of the visual judge or to claim that “evaluation on multimodal artifacts is not yet

mature” as a general fact. A larger human audit, ideally stratified by artifact type (e.g., 50+ per category), is necessary to validate the scoring reliability.

Finally, the **construction funnel** (Figure 2) reports a massive attrition rate (5,291 $\rightarrow$ 852). The paper describes the gates (length, fixture, network, self-containment) but does not provide a quantitative breakdown of rejection rates per gate. This omission makes it difficult to assess potential selection bias. If the majority of rejections are due to “ambiguous tasks” or “missing fixtures,” the resulting benchmark may systematically exclude the most complex, messy, or realistic enterprise scenarios, thereby inflating the reported performance of agents. A table detailing the number of instances rejected at each stage would significantly strengthen the evidence for the benchmark’s representativeness.

The main leaderboard results (Section 5.2) appear robust given the 32 combinations tested, but the supporting analyses for skill transfer and judge reliability need larger sample sizes to support the paper’s broader claims.

Action items.

- **[science]** The skill transfer experiment (Section 5.3) relies on a single task subclass (frontend page generation) with only 10 in-domain and 5 held-out tasks. This sample size is insufficient to support the broad claim that ‘skill injection is high variance’ or to generalize findings about creator-consumer fit. The authors must either expand the skill evaluation to multiple task classes or explicitly frame these results as a preliminary case study rather than a generalizable finding.
- **[science]** The human judge calibration study (Table 4) uses a sample of $n=48$ (24 text, 24 visual). While the text route shows reasonable correlation (Spearman 0.790), the visual route shows negative correlation. Given the paper’s heavy reliance on visual artifact evaluation, this sample size is too small to robustly characterize the failure mode of visual judges. A larger human audit (e.g., $n > 100$) stratified by artifact type is required to validate the scoring protocol.
- **[science]** The construction funnel (Figure 2) reports a reduction from 5,291 raw instances to 852 final tasks. The paper lacks a statistical breakdown of *why* instances were rejected at each gate (e.g., % rejected for missing fixtures vs. ambiguous prompts). Without this, it is impossible to assess if the final benchmark is biased toward ‘easy’ or ‘self-contained’ tasks, potentially inflating performance scores compared to the raw distribution of real workplace requests.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical analysis in this paper is generally descriptive and relies heavily on point estimates

and visual inspection of trends, which is common for benchmark papers but leaves gaps in rigor regarding significance and uncertainty.

First, the **Judge Reliability** analysis (Section 5.4, Table 2) presents a critical finding: the visual judge has a negative Spearman correlation ($\rho = -0.259$) with human raters on a sample of $n = 24$. While the direction of the correlation is clear, the paper fails to report a p-value or a confidence interval. Given the small sample size, it is statistically important to verify that this negative correlation is significantly different from zero. Without this, the claim that “evaluation on multimodal artifacts is not yet mature” rests on a potentially noisy estimate.

Second, the **Skill Evaluation** (Section 5.3) suffers from a lack of statistical power analysis or uncertainty quantification. The authors distill skills and test them on only **5 held-out tasks** per consumer. They report mean score deltas (e.g., +0.0681 for GPT-5.5, -0.0941 for Haiku 4.5) but provide no standard errors, confidence intervals, or significance tests. With $N = 5$, the standard error of the mean could be substantial, making it difficult to distinguish a true skill transfer effect from random variance in task difficulty. A paired statistical test (e.g., Wilcoxon signed-rank) comparing pre- and post-injection scores would be necessary to support the claim that specific creators are “strong” or “weak.”

Third, the **Cost-Score Trade-off** (Section 5.2) describes a “log-like shape” and “diminishing marginal returns” based on a scatter plot of 32 harness-model combinations. While the visual trend is plausible, the paper lacks a quantitative model fit. Reporting a correlation coefficient (Pearson or Spearman) or the R^2 of a fitted log-linear regression would provide the necessary statistical evidence to support the claim of a specific functional form rather than a general observation.

Finally, the **Scalability Check** (Section 5.2, Table 3) compares scores across the full set (852 tasks) but does not report confidence intervals for these aggregate scores. Given the heterogeneity of the task set, providing a 95% confidence interval for the mean score of each model would better contextualize the differences between models (e.g., is the 0.017 difference between GPT-5.5 and Sonnet 4.6 statistically significant?).

Overall, the paper presents a rich dataset but treats the resulting metrics as deterministic facts rather than estimates with uncertainty. Adding significance tests for the small-sample experiments and confidence intervals for the aggregate scores would significantly strengthen the statistical validity of the conclusions.

Action items.

- **[writing]** The statistical analysis in this paper is generally descriptive and relies heavily on point estimates and visual inspection of trends, which is common for benchmark papers but leaves gaps in rigor regarding significance and uncertainty. First, the Judge Reliability analysis (Section 5.4, Table 2) presents a critical finding: the visual judge has a negative Spearman correlation ($\rho = -0.259$) with human raters on a sample of $n = 24$. While the direction of the correlation is clear, the paper fa

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript presents a compelling benchmark with a clear narrative structure, effectively guiding the reader from the problem definition to the construction pipeline and experimental results. The prose is generally dense and technical, appropriate for the venue, and the logical flow between sections is sound. However, the writing quality is marred by several avoidable mechanical errors that detract from the overall polish.

Specifically, there are noticeable typos and spacing issues. In Section 3, the word “scenarios” is misspelled as “senarios.” In Section 4.1, a missing space after a comma (“repair,while”) and an awkward line break within a sentence disrupt the reading rhythm. Furthermore, the Appendix, which contains extensive bilingual case studies, exhibits inconsistent English grammar and capitalization in the translated portions (e.g., “generate a evaluation report”). While the core scientific argument is clear, these surface-level errors suggest the manuscript has not undergone a final, rigorous proofreading pass. Addressing these specific mechanical issues would significantly improve the readability and professional presentation of the paper.

Action items.

- **[writing]** In Section 3, Paragraph 2 (Construction pipeline), the sentence ‘In any other senarios, the instance is risk-marked...’ contains a spelling error (‘senarios’ should be ‘scenarios’). This is a basic proofreading issue that undermines the professional polish of the manuscript.
- **[writing]** In Section 4.1 (Harness–model interaction), the sentence ‘...multi-step repair,while Hermes more frequently...’ is missing a space after the comma. Additionally, the paragraph contains a line break in the middle of a sentence (‘...truncated before the artifact-writing loop is completed. As a result...’), which appears to be a formatting artifact from the source that disrupts the reading flow.
- **[writing]** Throughout the Appendix (e.g., ‘End-to-End Construction Case’), the English translations of Chinese prompts often lack consistent capitalization at the start of sentences or proper punctuation (e.g., ‘generate a evaluation report’ instead of ‘generate an evaluation report’). While these are translations, the inconsistency in the English text reduces readability and suggests a lack of final proofreading.